1 Interprocess communication

1.1 Producer–consumer buffer

> **Theorem 1.1.** *If access to the buffer is protected by mutual exclusion, the number of elements remains between zero and the capacity N.*

$$0 \leq n_{\text{buffer}} \leq N \tag{1.1}$$

Figure 1.1. Data flow through the shared buffer

Table 1.1. Buffer semaphore state

Semaphore	Initial value
`mutex`	1
`empty`	N
`full`	0

```python
def can_consume(items):
    return items > 0
```

Listing 1.1. Checking buffer availability

```
Input: A non-empty shared buffer
Output: The oldest element
wait(full)
wait(mutex)
x ← remove element
signal(mutex)
signal(empty)
return x
```

Algorithm 1.1. **Removing an element from the buffer**

The invariant in theorem 1.1, expressed by equation (1.1), prevents overflow and underflow. Figure 1.1 summarizes the flow, while table 1.1, listing 1.1, and algorithm 1.1 describe initialization and removal.